

MUHAMMAD DANISH

Mathematics Student • Data Analyst • AI & Automation Developer • Industrial Digitalization

Khanewal, Pakistan • +92 326 7370173 • muhammaddanish2327@gmail.com
Portfolio: muhammaddanish-sage.vercel.app GitHub: github.com/danishcorder
LinkedIn: linkedin.com/in/muhammad-danish-0b3727376

PROFESSIONAL SUMMARY

Mathematics undergraduate with a practical, evidence-based track record in data analytics, Python development, and AI/automation applied to industrial settings. During an 8-week internship at a spinning mill, moved from studying manual, Excel-based workflows to independently designing and building digital tools — including a computer-vision quality monitoring application and a role-based material requisition system — that replaced manual record-keeping with structured, automated processes. Works across data cleaning and analysis, backend APIs, and applied AI, with growing practical training in performance marketing analytics and a consistent focus on turning data into useful decisions and working software.

EDUCATION

Bachelor of Science — Mathematics

2023 – Present

Bahauddin Zakariya University, Multan

PROFESSIONAL EXPERIENCE

Industrial Digitalization Intern — Mahmood Group (Spinning Mill)

8-Week Internship, 2026

- Studied real industrial workflows across the spinning mill — quality testing, store/inventory management, and production reporting — to identify processes still run manually on paper or Excel.
- Analyzed manual data-entry and record-keeping processes and proposed digital solutions, then moved from concept to working prototypes using Python, web frameworks, and applied data analysis.
- Designed and developed TexGauge IQ, a computer-vision desktop application for sliver quality monitoring, as sole developer — see Projects for detail.
- Designed and developed a Digital Material Requisition & Approval System with role-based workflows (Worker, GM, Store) to replace manual requisition paperwork — see Projects for detail.
- Applied data cleaning and analysis techniques to ERP inventory data and production records to support further automation and reporting initiatives.

AI Developer, Data Analyst & Web Developer — Self-Directed

2023 – Present

- Built AI-powered applications and automation tools in Python, including an AI email assistant, an API key generator, and workflow automation utilities.
- Developed data analysis pipelines covering cleaning, exploratory data analysis, visualization, and reporting on industrial and academic datasets.
- Built responsive web applications (HTML, CSS, JavaScript) with MySQL-backed data storage, including a team webpage and a business website (Shangrila Garden).

Live demos: [AI Email Assistant](#) • [API Key Generator](#) • [Team Webpage](#) • [Shangrila Garden](#) • [Personal AI Assistant](#)

PERFORMANCE MARKETING DEVELOPMENT

Google & Meta Ads — Practical Training / Self-Directed

2026 – Present

- Developing practical capability in Google Ads and Meta Ads: campaign structure, keyword/audience targeting, bidding, budgets, and conversion-focused optimization.
- Applying an analytics-first approach to campaign metrics including impressions, clicks, CTR, CPC, conversions, CPA, and ROAS.
- Practicing campaign reporting, A/B testing concepts, audience/creative testing, and data-driven budget decisions using an existing foundation in Excel, Power BI, Python, and data analysis.

CORE TECHNICAL SKILLS

Programming	Python, JavaScript, HTML, CSS, SQL
Data Analytics	Pandas, NumPy, Matplotlib, Seaborn, Excel, Power BI, Tableau
AI & Automation	AI Agents, Prompt Engineering, Workflow Automation, AI Fundamentals
Backend & APIs	FastAPI, REST APIs, JWT Authentication, SQLite
Databases	MySQL, SQLite, SQL, DBMS
Tools & Platforms	Git, GitHub, Linux, Vercel, Microsoft Office

FEATURED INDUSTRIAL PROJECTS

Work progressed from observing the mill's manual processes, to identifying automation opportunities, to designing and building the digital tools below.

TexGauge IQ — Smart Sliver Quality Monitoring System

Python Computer Vision Serial Communication (RS232) Windows Desktop

PROBLEM Sliver testing relied on manual sample measurement, hand-recorded weights, manual calculations, and register- or Excel-based record-keeping — slow and prone to inconsistency.

OBJECTIVE Build a digital system to make sliver quality monitoring faster, more consistent, and easier for operators to use.

SOLUTION

- Records sample information digitally and reads sample weight via an RS232 serial connection, removing manual transcription.
- Calculates quality metrics — including G/Y% and CV% — automatically, reducing manual calculation.
- Provides a clear pass/fail quality indication and a structured digital workflow in place of paper registers, packaged as a Windows desktop application for the mill floor.

INDUSTRIAL VALUE

- ✓ Designed to cut the time operators spend on manual calculation and record-keeping for every sliver sample tested.
- ✓ Reduces the chance of manual entry/calculation errors and improves consistency and traceability of quality records.
- ✓ Creates a foundation for future quality analytics and further automation.

Digital Material Requisition & Approval System

Python FastAPI SQLite JWT Auth Vanilla JS

PROBLEM Material and spare-part requisitions were handled through manual documentation, repeated data entry, and back-and-forth communication between departments.

OBJECTIVE Develop a smarter digital workflow for submitting, approving, and tracking material requisitions across departments.

SOLUTION

- Built a multi-role, role-based approval workflow (Worker → GM → Store) with JWT-secured authentication, replacing manual sign-off paperwork.
- Structured a searchable inventory catalog of mill-specific spare parts so requisitions reference standardized items instead of free-text descriptions.
- Backend built with FastAPI and SQLite, with a lightweight vanilla-JS frontend for fast adoption by mill staff.

INDUSTRIAL VALUE

- ✓ Designed to cut the back-and-forth and paperwork involved in getting a requisition approved across departments.
- ✓ Improves accessibility and retrieval of requisition records versus paper-based tracking, and is designed to extend to additional departments.

ERP Inventory Master Data Cleaning System

Python · Pandas · Regex · Excel

Built a Python-based system to parse unstructured, free-text ERP item descriptions (e.g. ring travellers, bearings, and material codes such as NFC/MSHF) into structured, standardized fields. Applies domain-specific parsing rules for spinning-mill part nomenclature, turning inconsistent legacy inventory records into structured data suitable for reporting, search, and downstream automation.

Yarn Production Forecasting & Analytics

Python · Pandas · Machine Learning

Developing a forecasting pipeline for yarn production using a full year of real mill production data, covering data cleaning, exploratory analysis, and an early-stage machine learning forecasting workflow to support production planning.

ADDITIONAL PROJECTS

- Smart Autonomous Spinning Mill (Concept Architecture) - proposed AGV, LiDAR/QR navigation, and AI decision-support architecture.
- Smart Spinning Mill QMS (Python, Data Analysis, Excel Export) - multi-department quality tracking with live calculations and dashboards.
- Sales Performance Dashboard (Power BI, Excel, SQL) — interactive dashboard tracking KPIs and production metrics with automated reporting.
- House Price Prediction System (Python, Pandas, NumPy, Machine Learning) — feature engineering and predictive modeling to estimate housing prices.
- Personal Portfolio Website (HTML, CSS, JavaScript, GitHub, Vercel) — responsive portfolio deployed for cross-device performance.

Live demo: [Portfolio Website](#)

CERTIFICATIONS & LANGUAGES

Certifications

Python & AI — DigiSkills (2026) · Freelancing — DigiSkills (2024) · Data Analysis

Languages

English (Fluent) · Urdu (Fluent) · Chinese (Basic)